Assignment #3

2026-06-14

R Markdown

##Load the data and then inspect it

```
library(caret)
```

```
## Loading required package: ggplot2

## Loading required package: lattice
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'

## The following object is masked from 'package:ggplot2':
##
##     element
```

```
mydata <- read.csv("UniversalBank-1.csv")
head(mydata)
```

```
##   ID Age Experience Income ZIP.Code Family CCAvg Education Mortgage
## 1  1  25          1     49    91107      4   1.6         1        0
## 2  2  45         19     34    90089      3   1.5         1        0
## 3  3  39         15     11    94720      1   1.0         1        0
## 4  4  35          9    100    94112      1   2.7         2        0
## 5  5  35          8     45    91330      4   1.0         2        0
## 6  6  37         13     29    92121      4   0.4         2      155
##   Personal.Loan Securities.Account CD.Account Online CreditCard
## 1             0                  1          0      0          0
## 2             0                  1          0      0          0
## 3             0                  0          0      0          0
## 4             0                  0          0      0          0
## 5             0                  0          0      0          1
## 6             0                  0          0      1          0
```

```
dim(mydata)
```

```
## [1] 5000   14
```

##Splitting the data into 60% training and 40% validation using the createDataPartition function to get a stratified sample.

```
set.seed(1)
trainIndex <- createDataPartition(mydata$'Personal.Loan',p=.60,list=FALSE)
train <- mydata[trainIndex,]
valid <- mydata[-trainIndex,]
```

##Checking to see if split was successful

```
dim(train)
```

```
## [1] 3000   14
```

```
dim(valid)
```

```
## [1] 2000   14
```

##Part A, creating the initial pivot table

```
loan.table <- table(train$CreditCard,train$'Personal.Loan',train$Online)
loan.table
```

```
## , ,  = 0
##
##
##         0    1
##   0   805   79
##   1   332   30
##
## , ,  = 1
##
##
##         0    1
##   0  1119  119
##   1   469   47
```

##Part B: To find P(Loan=1 given CC=1,Online=1), we focus on the pivot table, specifically on customers who both held a bank credit card and used online banking services. In this group 47 accepted the loan and 469 did not out of a total of 516 customers. To calculate the probability we do 47/516=.0911 or 9.11%. This percentage represents the proability that a customer will accept a loan offer if they have a credit card and use online banking.

##Part C: This is where we create two seperate pivot tables. The first one being loan and online. The second one being loan and credit card.

##Table 1: Loan vs Online

```
loan_online <- table(train$'Personal.Loan',train$Online)
loan_online
```

```
##
##         0    1
##   0 1137 1588
##   1  109  166
```

##Table 2: Loan vs Credit card

```
loan_cc <- table(train$Personal.Loan,train$CreditCard)
loan_cc
```

```
##
##         0    1
##   0 1924  801
##   1  198   77
```

##Part D: This part in the assignment is asking to calculate six probabilities P(CC=1|Loan=1) P(Online=1|Loan=1) P(Loan=1) P(CC=1|Loan=0) P(Online=1|Loan=0) P(Loan=0)

##P(CC=1/Loan=1)=77/275=.2800 ##P(Online=1/Loan=1)=166/275=.6036 ##P(Loan=1)=275/3000=0.0917 ##P(CC=1/Loan=0)=801/2725=0.2940 ##P(Online=1/Loan=0)=1588/2725=.5828 ##P(Loan=0)=2725/3000=0.9083

these results are very interesting because credit card holders make up about 28% of loan accepters but they also make up 29.4% of non acceptors. Those numbers are very alike. Also, 60.4% of acceptors use online banking and 58.3% of non acceptors use online banking. Again, these numbers are very similiar. So far the conclusion is these predictors dont really provide us with much information about whether someone will accept a loan offer.

##Part E: In this part we are going to plug these probabilities into naive bayes formula

P(Loan=1)*P(CC=1/Loan=1)*P(Online=1/Loan=1) 0.0917*0.2800*0.6036=0.0155

P(Loan=0)*P(CC=1/Loan=0)*P(Online=1/Loan=0) 0.9083*0.2940*0.5828=0.1556

##P(Loan=1/CC=1,Online=1)=0.0155/0.0155+0.1556=0.0906 or 9.06%, so naives bayes estimate is 9.06%, If we compare this to part B which was 9.11%, we can conclude that there is only a 0.05 percentage point difference. The probability in part B is more accurate (9.11%) because its computed directly from the data observed. It does not depend on the independence assumption. Naive bayes probability of 9.06% is still very close to the actual profitability.

##Part G: Actually building the Naive Bayes Model in R

```
loan.nb <- naiveBayes(as.factor(Personal.Loan)~CreditCard+Online,data=train)
loan.nb
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
```

```
##          0          1
## 0.90833333 0.09166667
##
## Conditional probabilities:
##    CreditCard
## Y        [,1]      [,2]
##   0 0.293945 0.4556506
##   1 0.280000 0.4498175
##
##    Online
## Y         [,1]      [,2]
##   0 0.5827523 0.4931950
##   1 0.6036364 0.4900334
```

##So now we can see what probability the model assigns if a customer has a credit card and uses online banking. What is the probability of them accepting the load offer?

```r
new.customer <- data.frame(CreditCard=1,Online=1)
predict(loan.nb,new.customer,type="raw")
```

```
##              0          1
## [1,] 0.9108348 0.08916516
```

##The naive bayes model estimated 0.0892 which is very close to the manual naive bayes calculation of 0.0906 done in part E and the direct probability of 0.0911 in part B. This indicates that the model has performed as expected and is consistent.